3.1.2 Amount of substance: Sheet 3

Water of crystallisation, empirical formula and concentration · 16 marks · about 25 minutes

Key information

- Water of crystallisation: x = moles of H_2O ÷ moles of anhydrous salt.
- Empirical formula: divide each % by A_r, divide by the smallest, round to whole numbers. Molecular formula = (M_r ÷ empirical formula mass) × empirical formula.
- **Concentration (mol dm^{-3}) = moles ÷ volume (dm^3)**. Remember ionic compounds can give more than one of each ion.
- Dilution: moles stay the same, so $c_1V_1 = c_2V_2$; water added = final volume – starting volume.
- Back titration: moles of acid added – moles left over (from titration) = moles that reacted.

Section A: Self-quiz

01 Which can be both an empirical and molecular formula of a stable compound?

[1 mark]

A CH_2O
B P_4O_{10}
C NH_2
D CH_3

02 A solution of lead(II) chloride (M_r = 278.2) contains 1.08 g of $PbCl_2$ in 100 cm^3 of solution. In this solution, the lead(II) chloride is fully dissociated into ions. What is the concentration of chloride ions in this solution?

[1 mark]

A 3.88×10^{-3} mol dm^{-3}
B 7.76×10^{-3} mol dm^{-3}
C 3.88×10^{-2} mol dm^{-3}
D 7.76×10^{-2} mol dm^{-3}

03 A student is provided with 5.00 cm^3 of 1.00 mol dm^{-3} ammonia solution. The student was asked to prepare an ammonia solution with a concentration of 0.050 mol dm^{-3}. What volume of water should the student add?

[1 mark]

A 45.0 cm^3
B 95.0 cm^3
C 100 cm^3
D 995 cm^3

04 What is the concentration, in mol dm^{-3}, of ethanol if there are 80 mg of ethanol (M_r = 46.0) per 100 cm^3 of blood?

[1 mark]

A 0.00017
B 0.0017
C 0.017
D 1.7

05 A gas cylinder contains 5.0 kg of propane (C_3H_8).

How many propane molecules are in the cylinder? The Avogadro constant, L = 6.022×10^{23} mol^{-1}

[1 mark]

A 6.8×10^{22}

B 7.2×10^{22}

C 6.8×10^{25}

D 7.2×10^{25}

Section B: Exam questions

06 A student added 627 mg of hydrated sodium carbonate ($Na_2CO_3.xH_2O$) to 200 cm^3 of 0.250 mol dm^{-3} hydrochloric acid in a beaker and stirred the mixture.

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

After the reaction was complete, the resulting solution was transferred to a volumetric flask, made up to 250 cm^3 with deionised water and mixed thoroughly.

Several 25.0 cm^3 portions of the resulting solution were titrated with 0.150 mol dm^{-3} aqueous sodium hydroxide. The mean titre was 26.60 cm^3 of aqueous sodium hydroxide.

Calculate the value of x in $Na_2CO_3.xH_2O$

Show your working.

Give your answer as an integer.

[7 marks]

07 Analysis of a hydrocarbon Y shows that it contains 83.7% by mass of carbon.

Calculate the empirical formula of Y.

Use this empirical formula and the relative molecular mass of Y (M_r = 86.0) to calculate the molecular formula of Y.

[4 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**A**	Source: 7405/3 Jun 2018 Q05	1
02	**D**	Source: 7405/3 Jun 2017 Q26	1
03	**B**	Source: 7405/3 Jun 2017 Q30	1
04	**C**	Source: 7404/2 Jun 2019 Q10	1
05	**C**	Source: 7404/2 Jun 2018 Q21 (formula added)	1
06	M1 HCl added = 0.050 mol and NaOH used in titration = 3.99×10^{-3} mol M2 So moles that would be needed to neutralise total excess HCl = $3.99 \times 10^{-3} \times 10 = 3.99 \times 10^{-2}$ mol M3 Therefore moles of HCl reacted with the $Na_2CO_3.xH_2O$ = $0.050 - 3.99 \times 10^{-2} = 0.0101$ mol M4 So moles $Na_2CO_3.xH_2O = 0.0101 \div 2 = 5.05 \times 10^{-3}$ mol M5 Conversion of mg to g = 0.627 g M6 $xH_2O = 0.627 \div 5.05 \times 10^{-3} - 106.0 = 18(.16)$ M7 so x = 1	Source: 7405/1 Jun 2018 Q10.1 (equation added) Alternative M2/M3: divide moles HCl by 10 = 0.005 and $0.005 - 3.99 \times 10^{-3} = 0.00101$, then × 10 = 0.0101 Alternative M6/M7: mass $Na_2CO_3 = 5.05 \times 10^{-3} \times 106.0 = 0.5353$ g and mass $H_2O = 0.627 - 0.5353 = 0.0917$ g; $0.0917 \div 18.0 = 5.094 \times 10^{-3}$ so ratio 1 : 1.009, x = 1	7
07	M1 dividing %s by relative atomic masses: C = 83.7 ÷ 12(.0), H = 16.3 ÷ 1(.0) M2 converting (C : H 6.975 : 16.3) to 3 : 7 M3 empirical formula = C_3H_7 M4 molecular formula = C_6H_{14}	Source: 7404/2 Jun 2017 Q02.2 M1 & M2 are for working; M3 and M4 marked independently Formulae with no working cannot score M1 or M2 Alternative: M1 83.7% of 86 is 72; M2 72 ÷ 12 gives 6 C atoms	4